

# Laptop choice as a Typst PDF vignette

## The model, evaluated live

```
library(ahp)
ahpFile <- system.file("extdata", "laptop.ahp", package = "ahp")
laptopAhp <- Load(ahpFile)
Calculate(laptopAhp)
Analyze(laptopAhp)
```

|   |             | Weight | Chromebook | MacBook | ThinkPad | Inconsistency |
|---|-------------|--------|------------|---------|----------|---------------|
| 1 | Buy Laptop  | 100.0% | 44.7%      | 30.3%   | 24.9%    | 0.9%          |
| 2 | --Price     | 54.0%  | 34.7%      | 7.7%    | 11.6%    | 0.0%          |
| 3 | --Battery   | 29.7%  | 8.0%       | 12.0%   | 9.6%     | NA            |
| 4 | --Ecosystem | 16.3%  | 2.0%       | 10.6%   | 3.8%     | 0.4%          |

The Chromebook wins with a total weight of 44.7%, driven by Price (54.0% of the decision). The winning priorities per criterion are 34.7% (Price), 12.0% (Battery, MacBook) and 10.6% (Ecosystem, MacBook).

## The colored table, natively in Typst

`AnalyzeTable()` returns a `formattable` object, i.e. an HTML table, which Typst cannot render. `AnalyzeTinytable()` produces the same colored table as a `tinytable` object, rendered natively in each output format (HTML, LaTeX/PDF and Typst) — no screenshots needed:

```
AnalyzeTinytable(laptopAhp)
```

|                   | Weight | Chromebook | MacBook | ThinkPad | Inconsistency |
|-------------------|--------|------------|---------|----------|---------------|
| <b>Buy Laptop</b> | 100.0% | 44.7%      | 30.3%   | 24.9%    | 0.9%          |
| <b>Price</b>      | 54.0%  | 34.7%      | 7.7%    | 11.6%    | 0.0%          |
| <b>Battery</b>    | 29.7%  | 8.0%       | 12.0%   | 9.6%     | NA            |
| <b>Ecosystem</b>  | 16.3%  | 2.0%       | 10.6%   | 3.8%     | 0.4%          |

For wide tables that do not fit on a Typst page, shrink the font, e.g. `AnalyzeTinytable(carAhp, fontsize = 0.8)`. The only difference with `AnalyzeTable()` is that no warning icon is shown for high inconsistency (the background color is kept).

The screenshot approach (capture the `formattable` table to PNG with `webshot2` and embed the image) remains available and is documented in the “Choosing a laptop with AHP” vignette.

## The decision tree as a figure

`Visualize()` returns a DiagrammeR `grViz` widget (interactive HTML), which Typst cannot render either: used live, knitr prints a “PhantomJS not found” message and embeds a cropped, partial image. Screenshot it once and embed the PNG:

```
htmlwidgets::saveWidget(Visualize(laptopAhp), "laptop-tree.html",
  selfcontained = TRUE)
webshot2::webshot("laptop-tree.html", "laptop-tree.png",
  selector = ".grViz", delay = 3)
```

The `selector = ".grViz"` crops tightly to the tree (no whitespace) and `delay = 3` lets viz.js finish drawing before the capture:

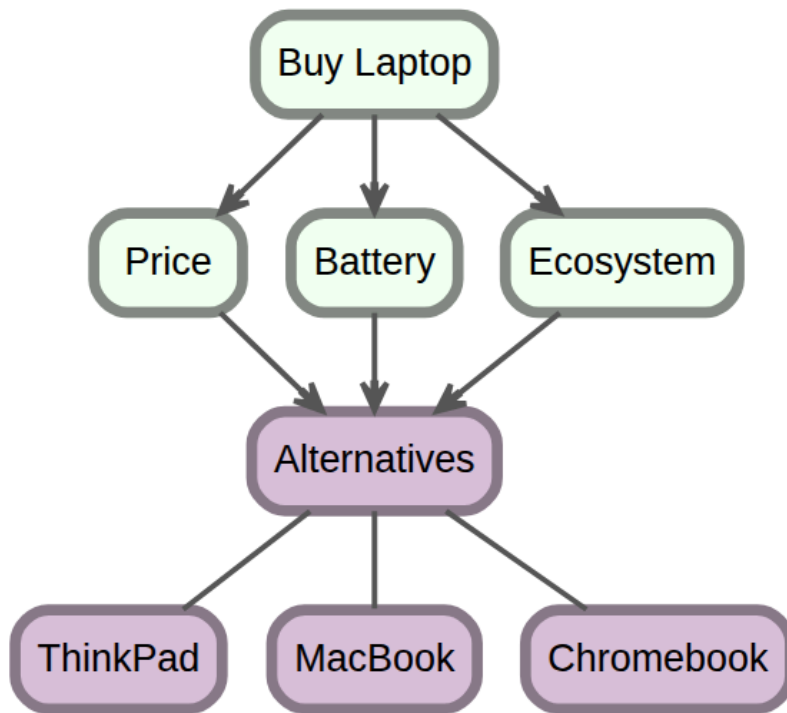


Figure 1: Decision tree